

JARVIS

Voice Assistant — Project Runbook

Personal desktop voice assistant with multilingual support

main.py • Python 3.8 • macOS

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1. Overview

JARVIS is a voice-controlled desktop assistant built in Python. It listens through the system microphone, transcribes speech to text via Google's speech recognition API, matches the transcribed text against a set of intents (time, jokes, opening apps, Wikipedia lookups, playing music, etc.), and replies out loud using text-to-speech.

The assistant supports two languages out of the box — English and Hindi — and can be extended to more. Language switching is explicit: the user says a phrase like "switch to hindi" rather than relying on automatic language detection, which proved unreliable for short voice commands during development.

Architecture at a glance

Stage	Library / Tool	Purpose
Speech-to-text	speech_recognition + Google Web Speech API	Converts microphone audio to text
Translation	deep-translator (Google Translate)	Translates non-English input to English for intent-matching, and English replies back to the active language
Text-to-speech (English)	pyttsx3 (nsss driver)	Fast, offline voice output for English
Text-to-speech (other languages)	gTTS + afplay	Network-based TTS with correct pronunciation for non-English replies
Knowledge lookups	wikipedia (Python package)	Answers "who is" / "what is" / "wikipedia" queries
System actions	subprocess + macOS "open"	Launches Calculator, TextEdit, Terminal, browser pages

2. Prerequisites & Environment Setup

2.1 System requirements

- macOS (the app uses macOS-specific commands: "open", "afplay", and the "nsss" pyttsx3 driver)
- Python 3.8 (running inside a conda environment named "jarvisssystem" in this project)
- A working microphone and speakers
- Active internet connection (required for speech recognition, Wikipedia, translation, and non-English text-to-speech)

2.2 Create and activate the environment

```
conda create -n jarvisssystem python=3.8 conda activate jarvisssystem
```

2.3 Install dependencies

```
pip install SpeechRecognition pyttsx3 wikipedia pyaudio pip install deep-translator gTTS
```

Note: If "pyaudio" fails to install via pip on macOS, install the system library first: brew install portaudio, then retry the pip install.

2.4 Grant microphone permission

The first time main.py runs, macOS will prompt for microphone access for Terminal (or your IDE). Approve this, or the assistant will silently fail to hear anything.

1. System Settings → Privacy & Security → Microphone
2. Enable the toggle for Terminal (or VS Code / your IDE)
3. Restart the terminal/IDE if the prompt was dismissed instead of approved

3. Running JARVIS

3.1 Start the assistant

```
conda activate jarvissystem cd jarvis-system python main.py
```

On startup, JARVIS greets you based on the time of day and announces it's ready, then enters a continuous listen loop:

```
JARVIS (en): Good Morning sir! How are you doing? JARVIS (en): I am JARVIS. Tell me  
sir how can i help you? Say 'switch to hindi' or 'switch to english' any time to  
change my language. Listening... Recognizing...
```

3.2 Stop the assistant

- Say "exit" — JARVIS says goodbye and the process ends cleanly
- Or press Ctrl+C in the terminal to force-stop at any time

3.3 Sample session output

A realistic terminal transcript covering a few commands in one session, included so you know what "working correctly" looks like without needing to run it first:

```
JARVIS: Good Morning sir! How are you doing? JARVIS: I am JARVIS. Tell me sir how  
can i help you? Listening... Recognizing... User said: what is your name what is  
your name JARVIS: My name is Adhok Listening... Recognizing... User said: tell me a  
joke tell me a joke JARVIS: Why don't programmers like nature? Too many bugs.  
Listening... Recognizing... none Listening... Recognizing... User said: who is  
Shahrukh Khan according to Wikipedia who is shahrukh khan according to wikipedia  
JARVIS: Searching wikipedia JARVIS: According to wikipedia JARVIS: Shah Rukh Khan  
(pronounced ['ʃɑ:hʊɪ xɑ:n] ; born as Shahrukh Khan on 2 November 1965), popularly  
known by the initials SRK, is an Indian actor and film producer renowned for his  
work in Hindi cinema. Referred to in the media as the "Baadshah of Bollywood" and  
"King Khan", he has appeared in more than 90 films and earned numerous accolades,  
including a National Film Award and 15 Filmfare Awards. Listening... Recognizing...  
User said: exit exit JARVIS: Good bye sir
```

Note: The blank "none" between the joke and the Wikipedia query is normal — it means one listen cycle didn't catch clear speech (silence or background noise) and the loop simply tried again.

3.4 Reading the terminal output

Every loop iteration prints diagnostic lines before speaking — use these to confirm what JARVIS actually heard and decided, independent of whether audio played correctly:

Line prefix	Meaning
Listening...	Microphone is open and capturing audio
Recognizing...	Audio captured; sending to Google Speech API
User said (<locale>):	Successful transcription and which STT locale was used
[lang: <code>]	The current active language and the translated-to-English query used for matching
JARVIS (<lang>):	The reply text and the language it will be spoken in
none	Recognition failed (silence, unclear audio, or network issue) — loop continues

4. Feature Reference

Say any of the following (English shown; Hindi equivalents work once in Hindi mode).

Say this	JARVIS does
"what time is it"	Speaks the current system time
"what is your name" / "who are you"	Replies with its name
"how are you"	Small talk reply
"who made you"	Small talk reply
"thank you"	Small talk reply
"play music"	Plays a random audio file from ~/Music/Music/Media
"play <song name>" (e.g. "play shape of you")	Fuzzy-matches the name against filenames and plays that file if found; otherwise plays a random song
"open calculator"	Opens the macOS Calculator app
"open notepad"	Opens TextEdit
"open google"	Opens google.com in the default browser
"open terminal" / "open cmd"	Opens a new Terminal window
"open calendar"	Opens Google Calendar in the browser
"tell me a joke"	Speaks a random programming joke
"open youtube <search term>"	Opens a YouTube search for the term
"open facebook"	Opens facebook.com
"who is <person>" / "what is <thing>" / "wikipedia <topic>"	Looks up a 2-sentence Wikipedia summary and reads it aloud
"switch to hindi"	Switches recognition + replies to Hindi from this point on
"switch to english"	Switches back to English
"exit"	Says goodbye and quits

5. Live Demo Script

A suggested run-through for demoing the project end to end, roughly 3–4 minutes.

Step 1 — Launch

4. Run `python main.py` and let the greeting play out loud.
5. Point out the terminal is printing what JARVIS is doing (Listening / Recognizing / JARVIS reply) — this is the transparency layer for the demo.

Step 2 — Core commands

6. Say "what time is it" — confirm it speaks the current time.
7. Say "tell me a joke" — confirm a random joke plays.
8. Say "what is your name" — confirm it says its name (and doesn't false-match on "time" logic).

Step 3 — Knowledge lookup

9. Say "who is Sundar Pichai" (or any well-known figure).
10. Point out the terminal shows the search term extracted, then the 2-sentence Wikipedia summary is read aloud.
11. Optionally demo a graceful failure: say "who is qwxyznonexistent" and show the "could not find anything on that topic" fallback instead of a crash.

Sample terminal output from an actual run ("who is Shahrukh Khan according to Wikipedia"):

```
User said: who is Shahrukh Khan according to Wikipedia who is shahrukh khan
according to wikipedia JARVIS: Searching wikipedia JARVIS: According to wikipedia
Shah Rukh Khan (pronounced ['ʃa:hʊɪ xä:n] ; born as Shahrukh Khan on 2 November
1965), popularly known by the initials SRK, is an Indian actor and film producer
renowned for his work in Hindi cinema. Referred to in the media as the "Baadshah of
Bollywood" and "King Khan", he has appeared in more than 90 films and earned
numerous accolades, including a National Film Award and 15 Filmfare Awards. JARVIS:
Shah Rukh Khan (pronounced ['ʃa:hʊɪ xä:n] ; born as Shahrukh Khan on 2 November
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renowned for his work in Hindi cinema. Referred to in the media as the "Baadshah of
Bollywood" and "King Khan", he has appeared in more than 90 films and earned
numerous accolades, including a National Film Award and 15 Filmfare Awards.
```

Note: The summary text is printed once (for the terminal/log) and then spoken via the "JARVIS:" line — that repetition is expected, not a bug. Also note the query matched on "who is", not the literal word "wikipedia" — this demonstrates the broadened trigger condition covered in section 4.

Step 4 — System actions

12. Say "open calculator" — Calculator app opens.
13. Say "play music" — a random song plays from the Music library.
14. Optionally say "play <a real song name from the library>" to show name-based matching.

Step 5 — Multilingual switch

15. Say "switch to hindi" — JARVIS confirms the switch (spoken in Hindi via gTTS).
16. Ask a question in Hindi, e.g. "tumhara naam kya hai" ("what is your name") — show the reply comes back in Hindi.
17. Say "switch to english" to return to English mode.

Step 6 — Exit

18.Say "exit" — JARVIS says goodbye and the process terminates.

6. Multilingual Mode Walkthrough

How language switching works

JARVIS does not attempt to auto-detect spoken language from short commands. Automatic detection (via libraries like `langdetect`) was tested during development and found unreliable on 2–4 word phrases — e.g. the word "exit" was misidentified as Catalan, and romanized Hindi ("tera naam kya hai") was misidentified as Indonesian. Instead, the active language is only ever changed by an explicit spoken command.

Say	Effect
"switch to hindi"	CURRENT_LANG becomes "hi". Recognition now uses the hi-IN locale; replies are translated to Hindi and spoken via gTTS.
"switch to english"	CURRENT_LANG becomes "en". Recognition uses en-IN; replies use the fast offline pyttsx3 voice.

Adding a new language

To support an additional language (e.g. Tamil or French), add one line to the `LANGUAGES` dictionary near the top of `main.py`:

```
LANGUAGES = {      "en": {"stt_locale": "en-IN", "switch_phrase": "switch to  
english"},      "hi": {"stt_locale": "hi-IN", "switch_phrase": "switch to hindi"},  
"ta": {"stt_locale": "ta-IN", "switch_phrase": "switch to tamil"}, }
```

No other code changes are required — translation and speech output both key off this dictionary automatically.

7. Troubleshooting

Symptom	Likely cause	Fix
No audio plays, but "JARVIS: ..." text prints fine	pytsx3's nsss driver going silent after repeated calls on the same engine instance	Already handled — <code>speak()</code> re-initializes a fresh pytsx3 engine on every call. If it recurs, test in isolation: <code>python3 -c "import pytsx3; e=pytsx3.init('nsss'); e.say('test'); e.runAndWait()"</code>
JSONDecodeError crash during Wikipedia lookup	Wikipedia API returned an empty/non-JSON response (network issue, disambiguation, or no matching page)	Already handled — wrapped in try/except for DisambiguationError, PageError, and general exceptions
"what is your name" doesn't reply, or replies incorrectly	Intent branch order or overly broad substring matching	Confirm "your name" / "who are you" branch appears before any broader catch-alls in the elif chain
Wikipedia doesn't trigger for "who is X"	Query didn't contain the literal word "wikipedia" and the phrase didn't match "who is"/"what is"	Confirm phrasing matches one of the three trigger conditions in the wikipedia elif branch
Hindi speech transcribed as garbled English text	en-IN locale was used instead of hi-IN	Say "switch to hindi" first — recognition locale only changes when CURRENT_LANG changes
Nothing happens after speaking ("none" printed)	Recognizer couldn't transcribe the audio — silence, background noise, or short phrase time limit	Speak clearly and promptly after "Listening..." appears; check mic input levels in System Settings
Translation fails / falls back to English text unexpectedly	No internet connection, or Google Translate temporarily unreachable	<code>translate_text()</code> catches this and returns the original text — check the logs/application.log file for the exact error
"play music" says folder not found	Music library path doesn't match ~/Music/Music/Media on this machine	Update the music_dir path inside play_music() to match your actual Apple Music library location

8. Known Limitations

- Only two languages are wired up by default (English, Hindi) — more require a one-line addition per language, plus a gTTS-supported language code.
- Language switching is manual only; there is no automatic detection of spoken language.
- Non-English text-to-speech (gTTS) requires an internet connection and has noticeably higher latency than the offline English voice.
- Wikipedia lookups return only a 2-sentence summary; no follow-up or "read more" capability.
- Music playback opens files via the macOS "open" command, which hands off to whatever app is set as the default handler (e.g. Music.app) — it does not have in-app playback controls (pause/skip) from JARVIS itself.
- Song name matching is a simple substring match — very short or ambiguous names may match unintended files.
- All "open <app>" commands are macOS-specific ("open", "-a" flag) and will not work unmodified on Windows or Linux.

9. File & Folder Reference

Path	Purpose
main.py	Entire application: setup, all intent logic, and the main listen loop
logs/application.log	Runtime log file — translation errors, speech errors, and other exceptions are logged here even when not printed to terminal
~/Music/Music/Media	Source folder for the "play music" / "play <song>" feature

End of runbook.